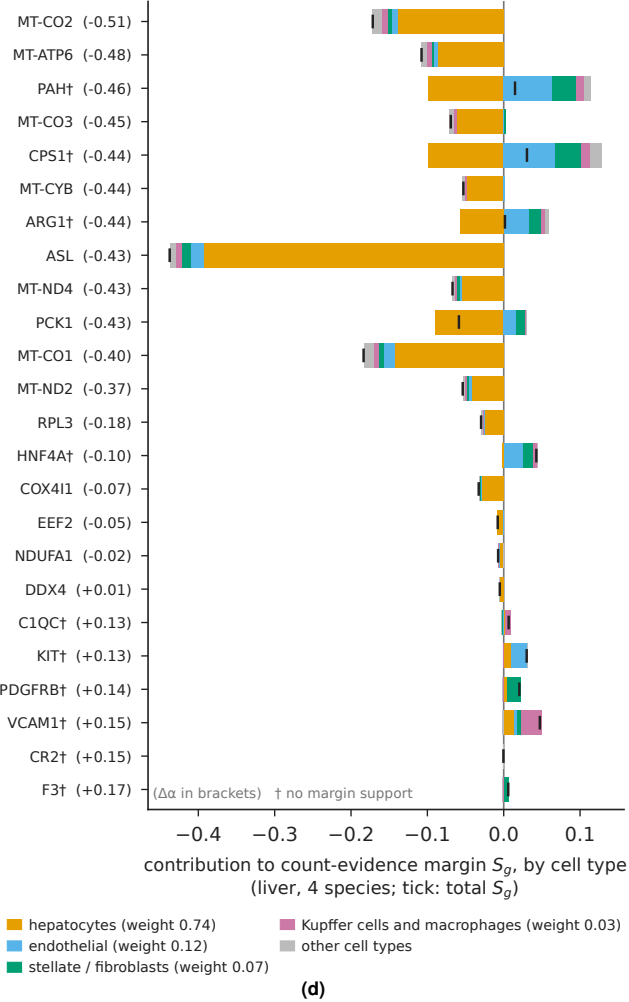
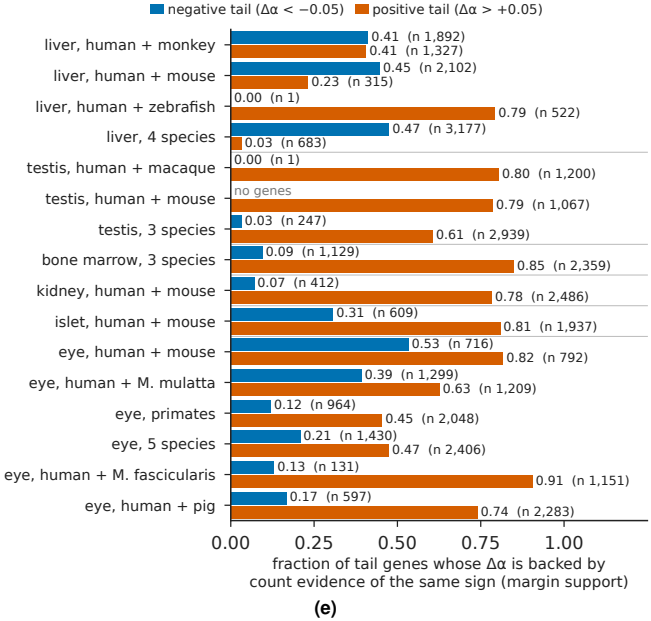
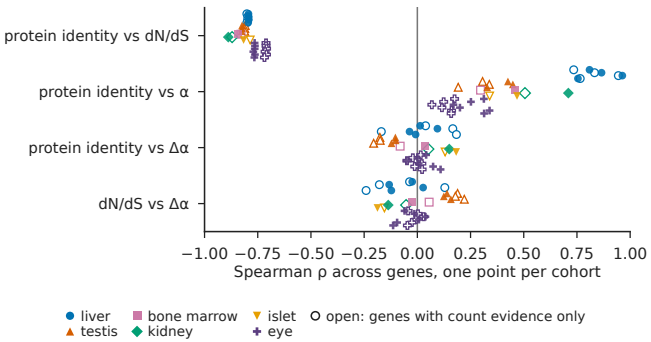

MT-CO2 (-0.51)
MT-ATP6 (-0.48)
PAH† (-0.46)
MT-CO3 (-0.45)
CPS1† (-0.44)
MT-CYB (-0.44)
ARG1† (-0.44)
ASL (-0.43)
MT-ND4 (-0.43)
PCK1 (-0.43)
MT-CO1 (-0.40)
MT-ND2 (-0.37)
RPL3 (-0.18)
HNF4A† (-0.10)
COX4I1 (-0.07)
EEF2 (-0.05)
NDUFA1 (-0.02)
DDX4 (+0.01)
C1QC† (+0.13)
KIT† (+0.13)
PDGFRB† (+0.14)
VCAM1† (+0.15)
CR2† (+0.15)
F3† (+0.17)
(Δα in brackets) † no margin support
−0.4 −0.3 −0.2 −0.1 0.0 0.1
contribution to count-evidence margin S_g, by cell type
(liver, 4 species; tick: total S_g)
hepatocytes (weight 0.74)
endothelial (weight 0.12)
stellate / fibroblasts (weight 0.07)
Kupffer cells and macrophages (weight 0.03)
other cell types

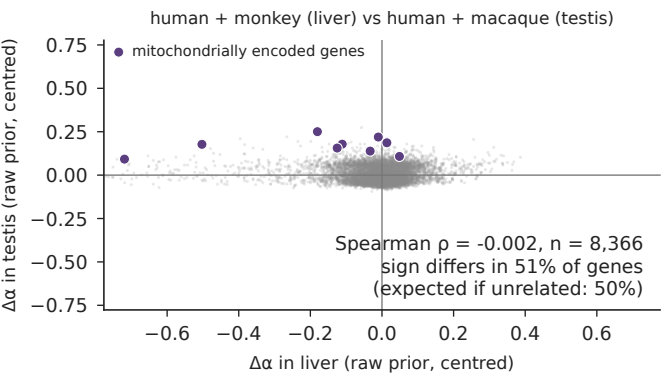
(d)


negative tail (Δα < −0.05)
positive tail (Δα > +0.05)
liver, human + monkey 0.41 (n 1,892) 0.41 (n 1,327)
liver, human + mouse 0.45 (n 2,102) 0.23 (n 315)
liver, human + zebrafish 0.00 (n 1) 0.79 (n 522)
liver, 4 species 0.47 (n 3,177) 0.03 (n 683)
testis, human + macaque 0.00 (n 1) 0.80 (n 1,200)
testis, human + mouse no genes 0.79 (n 1,067)
testis, 3 species 0.03 (n 247) 0.61 (n 2,939)
bone marrow, 3 species 0.09 (n 1,129) 0.85 (n 2,359)
kidney, human + mouse 0.07 (n 412) 0.78 (n 2,486)
islet, human + mouse 0.31 (n 609) 0.81 (n 1,937)
eye, human + mouse 0.53 (n 716) 0.82 (n 792)
eye, human + M. mulatta 0.39 (n 1,299) 0.63 (n 1,209)
eye, primates 0.12 (n 964) 0.45 (n 2,048)
eye, 5 species 0.21 (n 1,430) 0.47 (n 2,406)
eye, human + M. fascicularis 0.13 (n 131) 0.91 (n 1,151)
eye, human + pig 0.17 (n 597) 0.74 (n 2,283)
0.00 0.25 0.50 0.75 1.00
fraction of tail genes whose Δα is backed by
count evidence of the same sign (margin support)

(e)


protein identity vs dN/dS
protein identity vs α
protein identity vs Δα
dN/dS vs Δα
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
Spearman ρ across genes, one point per cohort
liver
bone marrow
islet
testis
kidney
eye
open: genes with count evidence only

(g)


human + monkey (liver) vs human + macaque (testis)
mitochondrially encoded genes
Δα in testis (raw prior, centred)
0.75 0.50 0.25 0.00 −0.25 −0.50 −0.75
Spearman ρ = -0.002, n = 8,366
sign differs in 51% of genes
(expected if unrelated: 50%)
−0.6 −0.4 −0.2 0.0 0.2 0.4 0.6
Δα in liver (raw prior, centred)
human + mouse, liver vs testis
0.4 0.2 0.0 −0.2 −0.4
Spearman ρ = -0.260, n = 8,233
sign differs in 60% of genes
(expected if unrelated: 50%)
−0.4 −0.2 0.0 0.2 0.4
Δα in liver (raw prior, centred)

(f)